

YOUR NAME
YOUR EMAIL
October 6, 2026

Homework 3

Don't forget an academic integrity statement (see syllabus for more details)!

1. Finish problems 2–3, then complete the academic integrity statement.
2. **Building off of the previous homework** In Homework 2, you should have performed an ANOVA on the dataset in the `rats.csv` file. If done correctly, you would have found a statistically significant treatment effect (at the $\alpha = 0.05$ level), with a P-value of 0.0102. That is, there is *some* evidence of differences between the treatment means.

We now would like to expand on these results.

- (a) What is the estimated treatment mean μ_i for each of the treatments?
 - (b) If you wanted to compare all treatment means against each-other (all pairwise comparisons), how many comparisons would you be making?
 - (c) Using an $\alpha = 0.05$ significance level, perform all pair-wise comparisons using Tukey's-HSD adjustment, and list the adjusted P-value for each comparison. What are the significant differences you found?
 - (d) **(For 6000-level students):** Now perform the same pair-wise comparisons without making the Tukey adjustment. Does it change the outcomes?
- 3.